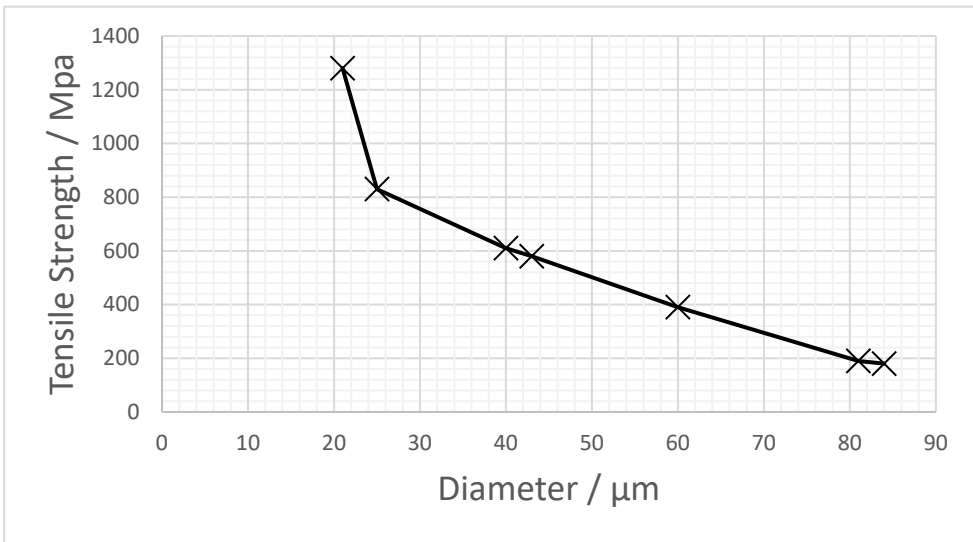


| Question Number | Answer                                                                                                                                                                      | Additional Guidance                                                                                                                                                                                                             | Mark       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>2(a)(i)</b>  | <p>An answer that includes the following steps:</p> <ul style="list-style-type: none"> <li>• reading of diameter using scale (1)</li> <li>• width calculated (1)</li> </ul> | <p>Example of calculation:</p> <p>19 units</p> <p>So actual width is <math>19 \times 3 = 57 \text{ (}\mu\text{m)}</math></p> <p>correct answer with no working gains both marks</p> <p>Accept diameters from 15 to 20 units</p> | <b>(2)</b> |

| Question Number | Answer                                                                                                                                          | Additional Guidance                   | Mark       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------|
| <b>2(a)(ii)</b> | <p>An answer that includes the following points :</p> <ul style="list-style-type: none"> <li>• xylem (1)</li> <li>• sclerenchyma (1)</li> </ul> | <p>Mark first answer on each line</p> | <b>(2)</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                      | Additional Guidance                                                                                                                                               | Mark       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>2(b)(i)</b>  | <ul style="list-style-type: none"> <li>• fibre variables taken into account (1)</li> <li>• appropriate environmental variable controlled (1)</li> <li>• appropriate description of how masses are added (1)</li> <li>• adding masses until fibre breaks / measure the mass that {breaks the fibre / that the fibre can hold before breaking} (1)</li> </ul> | <p>e.g. length, age, mass, hydration level, plant extracted from</p> <p>e.g. temperature, humidity, distance between stands</p> <p>e.g. one at a time, gently</p> | <b>(4)</b> |

| Question Number          | Answer                                                                                                                                                                                                                                                                 | Additional Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Mark                     |                        |    |      |    |     |    |     |    |     |    |     |    |     |    |     |     |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------|----|------|----|-----|----|-----|----|-----|----|-----|----|-----|----|-----|-----|
| 2(b)(ii)                 | <p>A graph showing the following features:</p> <p>A axes correct (x - diameter, y - tensile strength) (1)</p> <p>L axes correctly labelled and with units (1)</p> <p>P correct plotting on a linear scale on y (1)</p> <p>S points joined with straight lines. (1)</p> |  <table><caption>Data points from the graph</caption><thead><tr><th>Diameter / <math>\mu\text{m}</math></th><th>Tensile Strength / Mpa</th></tr></thead><tbody><tr><td>20</td><td>1300</td></tr><tr><td>25</td><td>850</td></tr><tr><td>40</td><td>600</td></tr><tr><td>42</td><td>580</td></tr><tr><td>60</td><td>400</td></tr><tr><td>80</td><td>200</td></tr><tr><td>85</td><td>180</td></tr></tbody></table> | Diameter / $\mu\text{m}$ | Tensile Strength / Mpa | 20 | 1300 | 25 | 850 | 40 | 600 | 42 | 580 | 60 | 400 | 80 | 200 | 85 | 180 | (4) |
| Diameter / $\mu\text{m}$ | Tensile Strength / Mpa                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                          |                        |    |      |    |     |    |     |    |     |    |     |    |     |    |     |     |
| 20                       | 1300                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                          |                        |    |      |    |     |    |     |    |     |    |     |    |     |    |     |     |
| 25                       | 850                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                          |                        |    |      |    |     |    |     |    |     |    |     |    |     |    |     |     |
| 40                       | 600                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                          |                        |    |      |    |     |    |     |    |     |    |     |    |     |    |     |     |
| 42                       | 580                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                          |                        |    |      |    |     |    |     |    |     |    |     |    |     |    |     |     |
| 60                       | 400                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                          |                        |    |      |    |     |    |     |    |     |    |     |    |     |    |     |     |
| 80                       | 200                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                          |                        |    |      |    |     |    |     |    |     |    |     |    |     |    |     |     |
| 85                       | 180                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                          |                        |    |      |    |     |    |     |    |     |    |     |    |     |    |     |     |

| Question Number  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Additional Guidance   | Mark       |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------|
| <b>2(b)(iii)</b> | <p>An answer including three of the following points :</p> <ul style="list-style-type: none"> <li>• the larger the diameter the less tensile strength / negative correlation (1)</li> <li>• non-linear (1)</li> <li>• steepest decrease {between 21 <math>\mu\text{m}</math> and 25 <math>\mu\text{m}</math> / up to 25 <math>\mu\text{m}</math>} (1)</li> <li>• less steep from {25-81 <math>\mu\text{m}</math> / 81-84 <math>\mu\text{m}</math> } (1)</li> </ul> | <b>Accept</b> reverse | <b>(3)</b> |